DSP in VLSI

Homework 2

Digital Filter

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Development Environment

HDL	Verilog HDL
Simulation Tools	Synopsys VCS and Verdi
Synthesis Tool	Synopsys DC Compiler
Process	TSMC 16nm(ADFP)
Verification	Python

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Step 1 — Filter Analysis

Please use Matlab/Python to draw the **impulse response** and **frequency response** (containing magnitude and phase) of the 25-tap sinc-based FIR filter. Use dB scale for magnitude and radian for phase versus normalized frequency. The x-axis must be marked with the correct label. **(20%)** Please explain whether the filter is high-pass, band-pass or low-pass and why. **(5%)**

1.1 Impulse Response

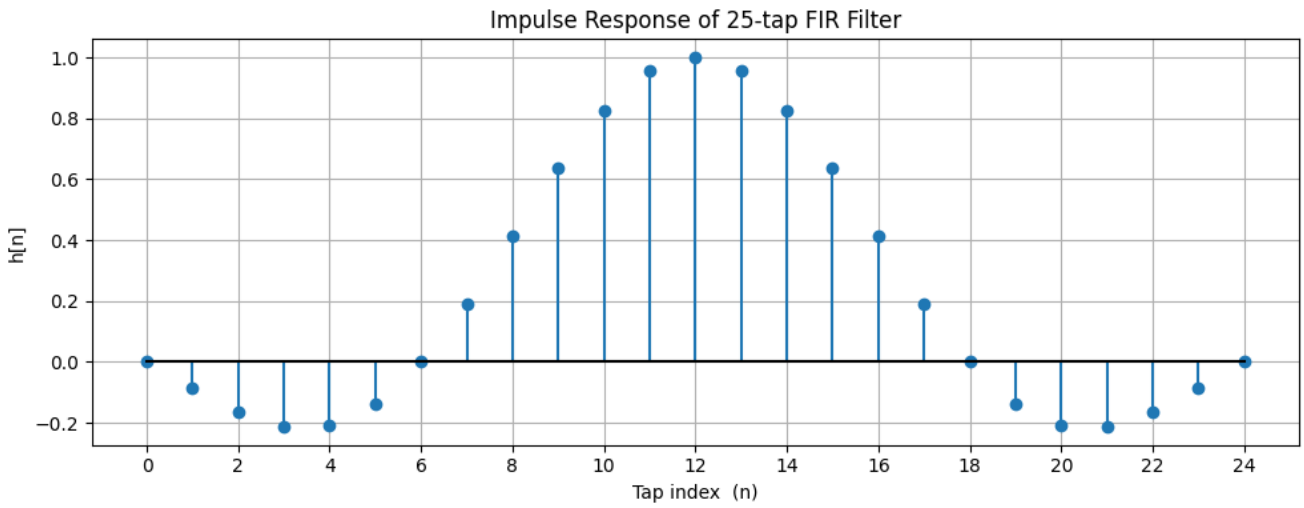

Figure 1.1: Impulse Response Diagram

1.2 Frequency Response and Filter Type

This is a low-pass filter. The filter coefficients are derived from the Sinc function, which corresponds to a rectangular frequency response (ideal low-pass filter) in the frequency domain. From the magnitude response, the passband is located at low frequencies ($\omega < 0.141\pi$) with approximately 16 dB gain, and the stopband attenuates high-frequency components by 20 to 40 dB. The 3dB cutoff frequency is at $\omega = 0.141\pi$, close to the theoretical value of$\pi/6$. The phase response is linear in the passband, consistent with a symmetric-coefficient FIR filter.

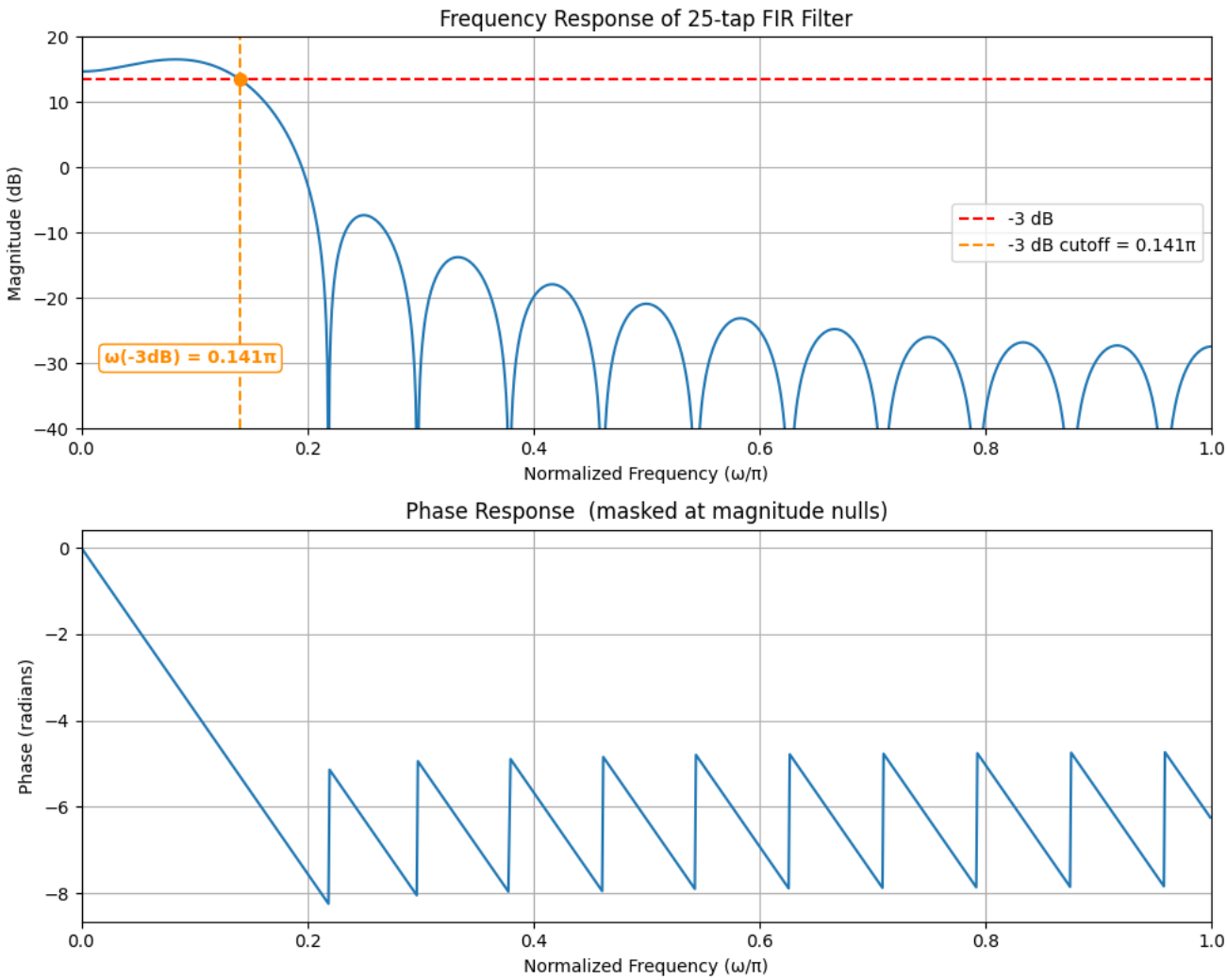

Figure 1.2: Frequency Response Diagram

Step 2 - Input and output waveforms

Draw the time-domain input and output waveforms after feeding 144 input samples $x[n] = \sin(-\frac{2\pi n}{128}) - \cos(\frac{2\pi n}{4})$, $0 \leq n \leq 143$, and observe the filtering effect. **(20%)**

2.1 Sinc Function Implementation

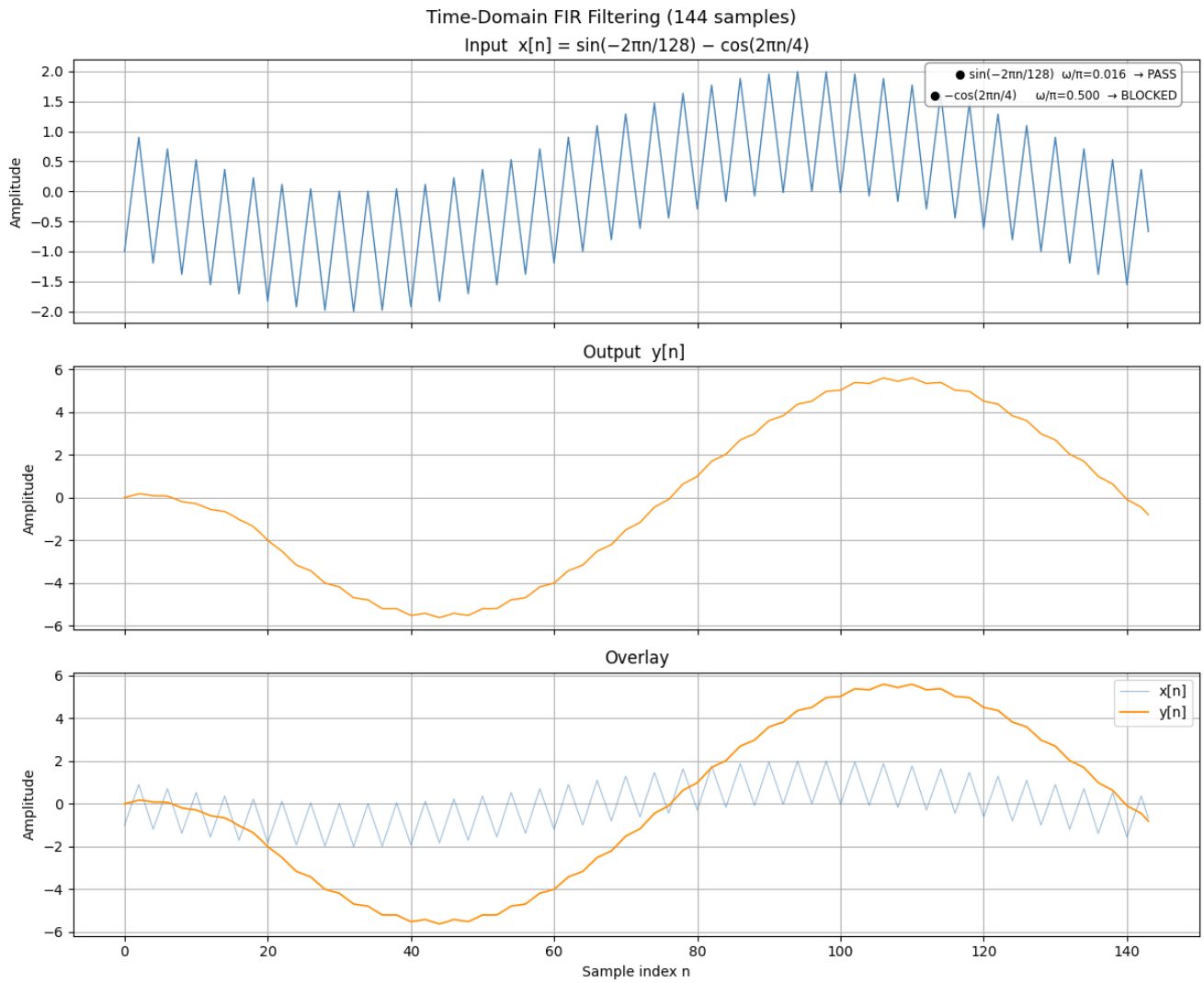

Figure 2.1: Time-Domain Input/Output Waveforms

Step 3 — Quantization

Please use **direct form** to implement the 25-tap FIR filter. Decide the word-lengths of each data path after additions and multiplications so that the RMSE error between the fixed-point output and the floating-point output is less than 2^{-a}. If $p = 0$, $a = 12$; otherwise $a = 11$, where $p = \mathrm{mod}(d_1, 2)$ and d_1 is the last digit of your student ID.

To show how you determine the word-length, scan the quantization error versus word-lengths from 9 bits to 20 bits and mark the final word-length settings in the block diagram. **(20%)**

Draw the fixed-point output of the time-domain signal and compare to the floating-point output. **(10% + 10%)** Show the frequency response of the fixed-point direct-form FIR filter and compare the magnitude error in passband (within 3 dB bandwidth), expressed in dB. **(10% + 5%)**

3.1 Word-Length Determination

Based on the analysis of fractional word lengths ranging from **9 to 20 bits**, the minimum word length required to meet the output error threshold was identified. The experimental results indicate that **the input requires a 13-bit fractional length**. With the input fixed, **the coefficients require 15 bits**. Following these assignments, **the multiplication results require 17 bits**, and finally, **the accumulation requires 17 bits**.

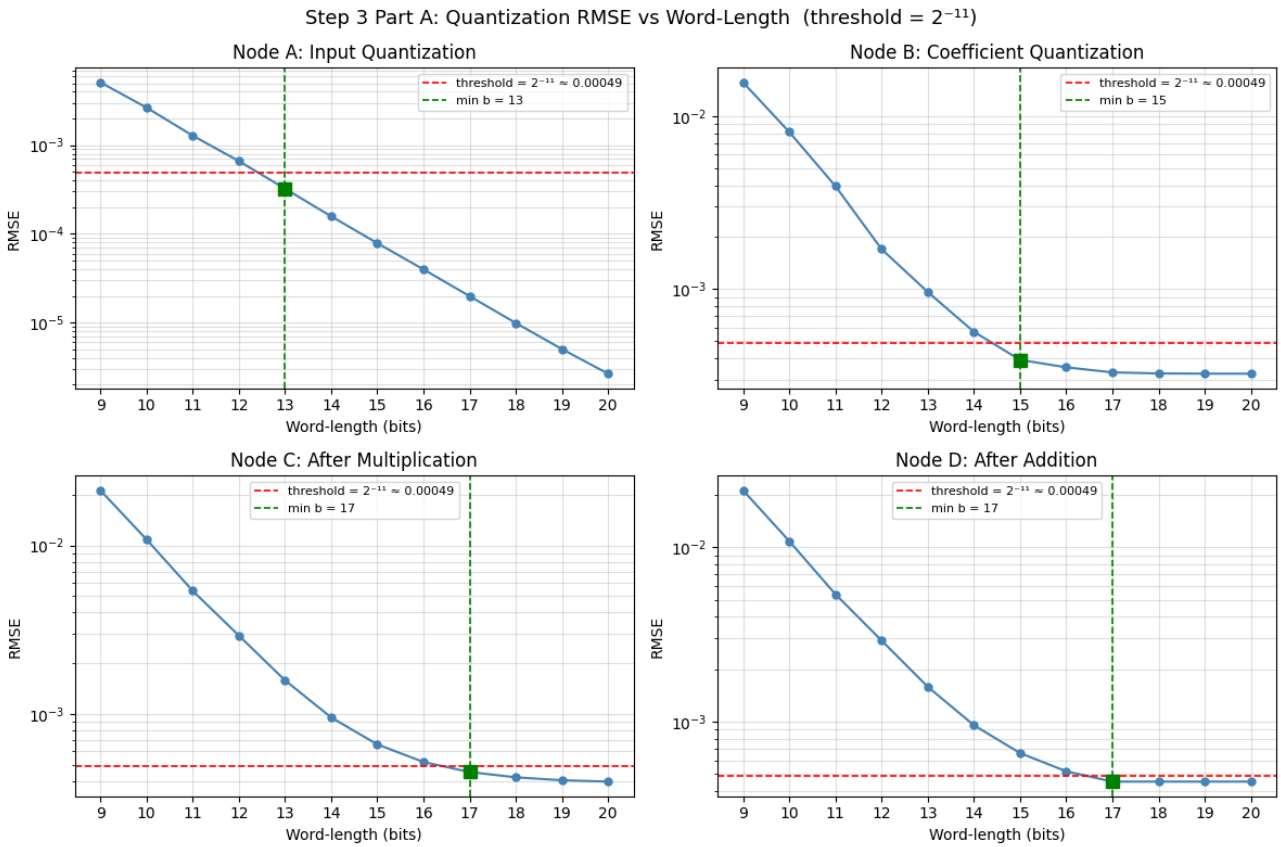

Figure 3.1: Scan the quantization error versus word-lengths from 9 bits to 20 bits

Furthermore, by incorporating the maximum integer word lengths for each stage, the complete fixed-point configurations are determined as follows:

Stage	Fractional Bits (F)	Total Word Length (W)
Input	13 bits	16 bits
Coefficient	15 bits	17 bits
Multiplication	17 bits	20 bits
Accumulation	17 bits	21 bits

Table 3.1: Word-length configuration for each stage.

3.2 Fixed-Point FIR Analysis

Based on the experimental results, the overall RMSE is 0.00045324, which is below the error threshold $2^{-11} = 0.00048828$. The maximum error in the passband is 0.00054532 dB.

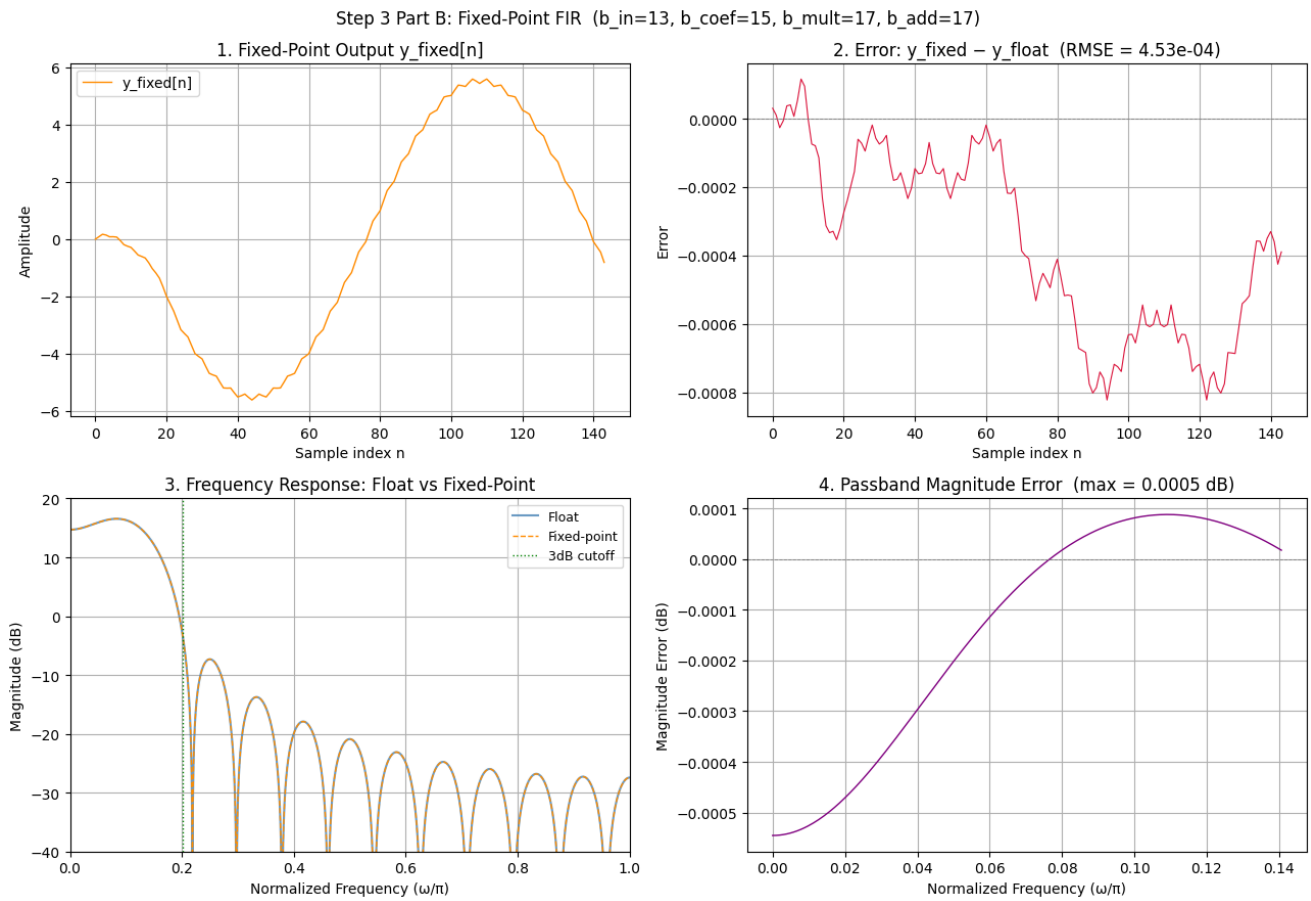

Figure 3.2: Fixed-Point FIR Analysis

Step 4 — Direct Form Implementation & Synthesis

Print out the behavior timing diagram for the first 20 samples and the last 20 samples **(10%)**. Show the errors of hardware outputs and Matlab floating-point results of direct form FIR filter output $y[n]$ for $0 \leq n \leq 143$ by figures. **The latency is not constrained. (10%)**

Paste the timing report about the critical path. Show the numbers of adders and multipliers in the critical path **(5%)**. List your timing constraint to achieve the highest operating frequency **(5%)**. Print out the **post-SYNTHESIS** timing diagram for the first 20 samples and the last 20 samples **(10%)**. and show the errors of post-synthesis simulation outputs and Matlab floating-point results of direct-form FIR filter output $y[n]$ for $0 \leq n \leq 143$ **(10%)**.

4.1 Behavior Simulation and Error Analysis

In this design, **a pipeline stage is incorporated every five taps,** which effectively reduces the depth of combinational logic, thereby improving the maximum clock frequency.

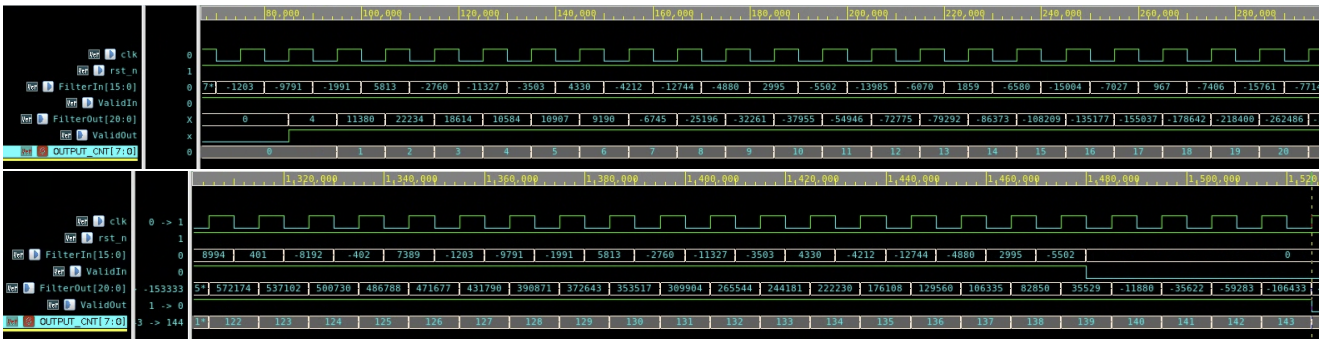

Figure 4.1: Behavior Timing Diagram (First 20 & Last 20)

In the testbench, the filter output is written to **output.dat**. The output is then converted from Q4.17 fixed-point to FP32 using Python, and compared against the original floating-point results for error analysis. The resulting RMSE is only 0.00045324, which is below the error threshold.

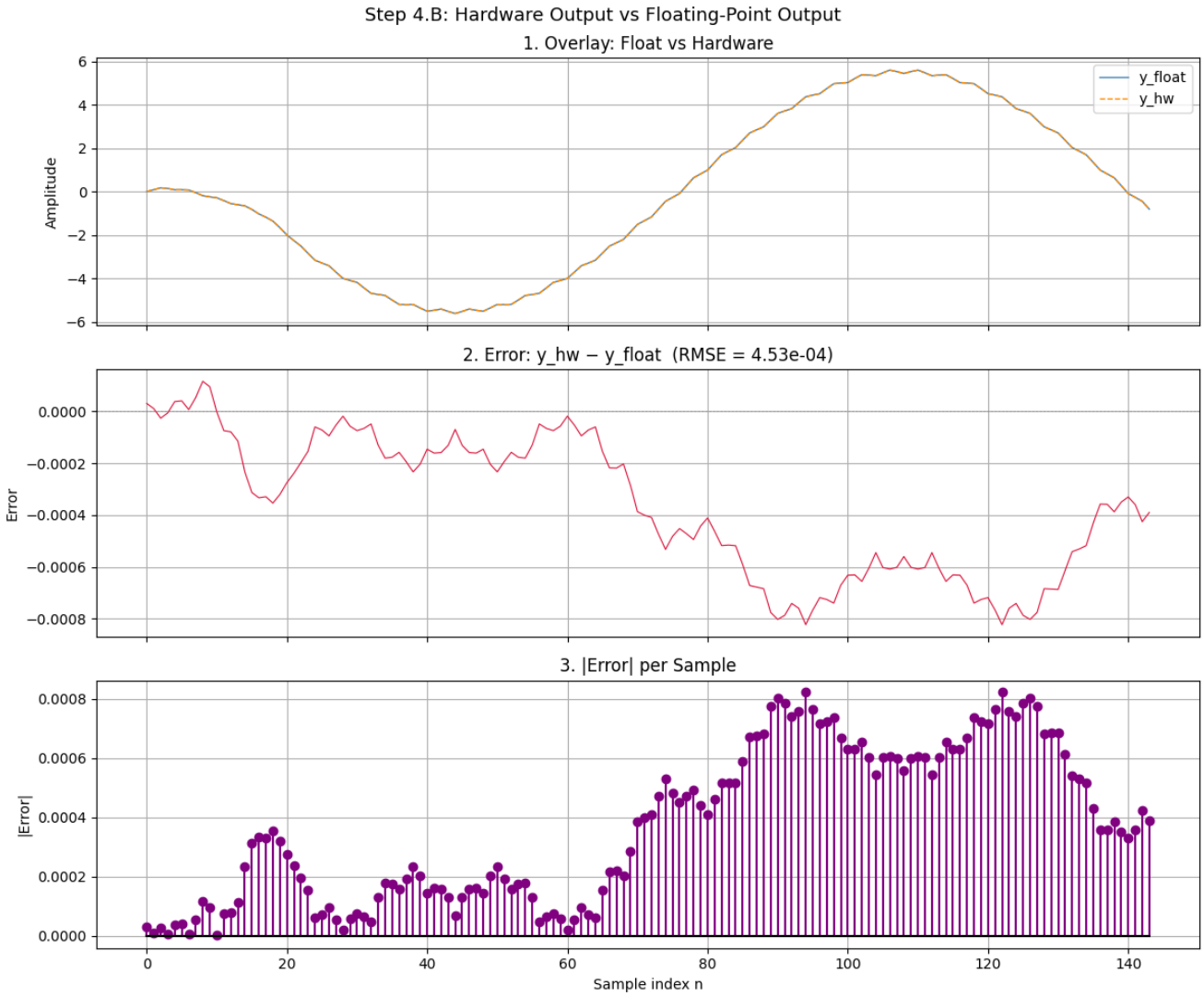

Figure 4.2: The errors of hardware outputs and Python floating-point result

4.2 Critical Path Report and Timing Constraint

The critical path of the direct-form FIR filter consists of **1 multiplier and 4 adders (Non-Optimization).** The maximum delay from the synthesis timing report is 0.90 ns and **the maximum operation frequency is 1GHz**.

```
Digital_Filter     ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs

Point                                                    Incr       Path
-----------------------------------------------------------------------
clock clk (rise edge)                                    0.00       0.00
clock network delay (ideal)                              0.50       0.50
X_D_reg[22][12]/CP (DFCNQD2BWP16P90)                     0.00       0.50 r
X_D_reg[22][12]/Q (DFCNQD2BWP16P90)                      0.04       0.54 r
U1768/ZN (CKND2BWP20P90)                                 0.01       0.55 f
U1641/CO (FA1D1BWP16P90LVT)                              0.03       0.58 f
U1732/S (FA1D1BWP20P90)                                  0.04       0.62 r
U1804/S (FA1D1BWP20P90)                                  0.04       0.66 f
U1854/S (FA1D1BWP20P90)                                  0.04       0.70 r
U1888/CO (FA1D1BWP20P90)                                 0.02       0.72 r
U1892/S (FA1D1BWP20P90)                                  0.04       0.76 f
U1878/S (FA1D1BWP16P90)                                  0.04       0.80 r
U448/ZN (NR2D1BWP16P90)                                  0.01       0.81 f
U442/ZN (INVD1BWP16P90)                                  0.01       0.82 r
U469/ZN (ND2D1BWP16P90LVT)                               0.01       0.83 f
U468/ZN (NR2D1BWP16P90LVT)                               0.01       0.84 r
U1899/ZN (ND2D1BWP20P90)                                 0.01       0.85 f
U620/ZN (NR2D1BWP20P90)                                  0.01       0.86 r
U4044/ZN (AOI21D1BWP16P90)                               0.01       0.87 f
U575/Z (XOR2D2BWP16P90LVT)                               0.02       0.88 r
U572/Z (BUFFD12BWP16P90LVT)                              0.02       0.90 r
FilterOut[18] (out)                                      0.00       0.90 r
data arrival time                                                   0.90

clock clk (rise edge)                                    1.00       1.00
clock network delay (ideal)                              0.50       1.50
clock uncertainty                                       -0.10       1.40
output external delay                                   -0.50       0.90
data required time                                                  0.90
-----------------------------------------------------------------------
data required time                                                  0.90
data arrival time                                                  -0.90
-----------------------------------------------------------------------
slack (MET)                                                         0.00
```

Figure 4.3: Timing Report

Parameter	Value (ns)
Clock Period	1
Clock Uncertainty	0.1
Clock Latency	0.1
Input / Output Maximum Delay	0.5

Table 4.1: Timing constraint settings.

4.3 Post-Synthesis Simulation and Error Analysis

Actually, the RMSE of Post-Synthesis is same as Behavior Simulation in Section 3.1 (only 0.00045324).

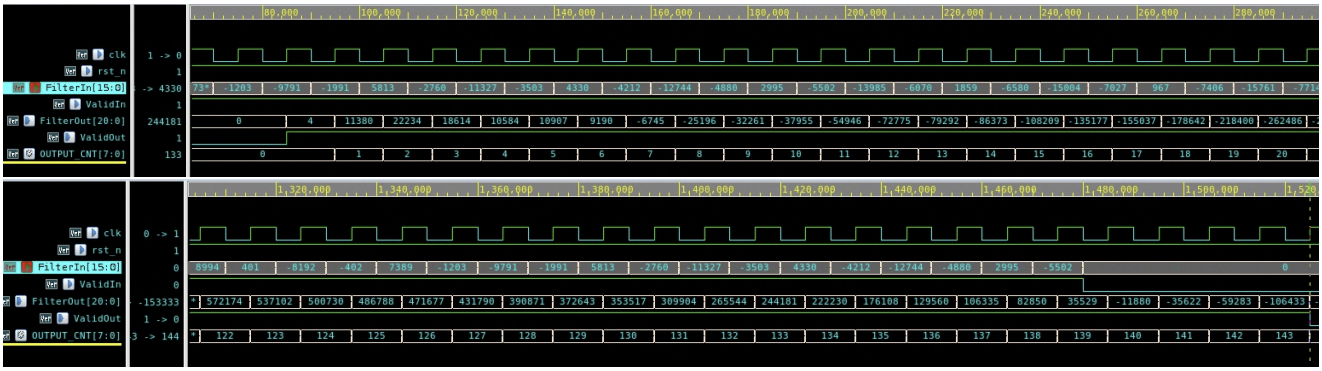

Figure 4.4: Post-Synthesis Timing Diagram (First 20 & Last 20)

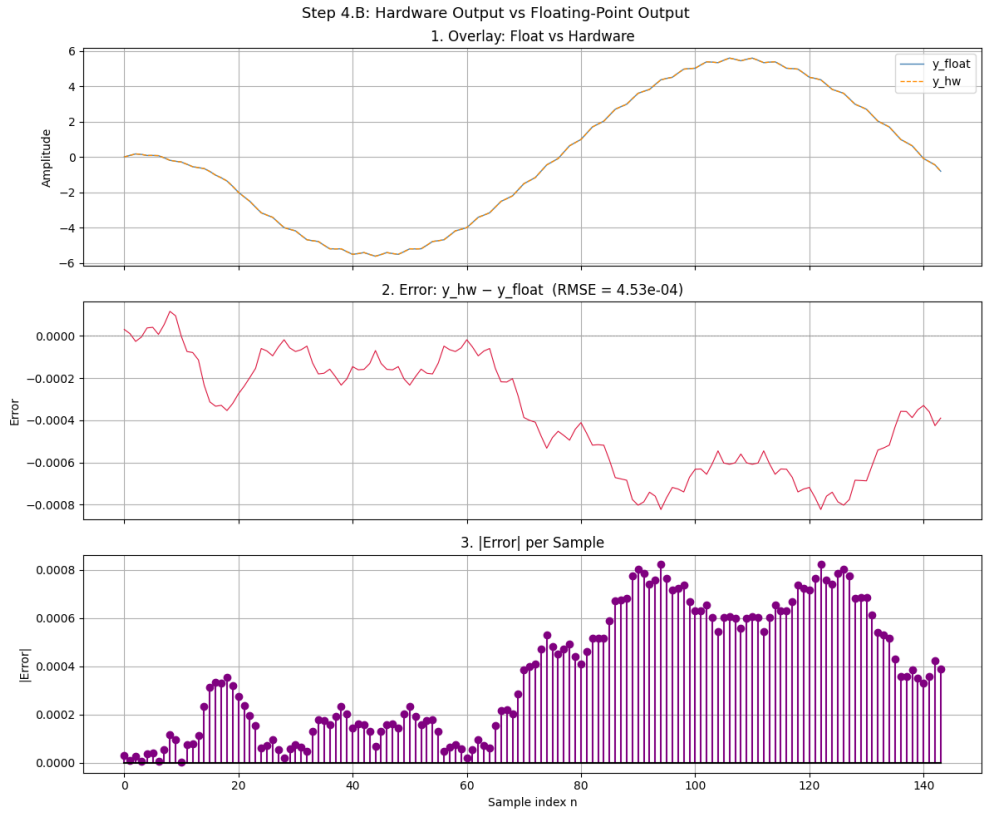

Figure 4.5: The errors of Post-Synthesis hardware outputs and Python floating-point result

Step 5 — 2X Speedup via Parallel Processing

Show that $y[0]$ **to** $y[143]$ **can be generated within** $T_{\text{out}}/2$ in the behavior timing diagram. Draw the block diagram and indicate your critical path **(15%)**. Show your synthesis report of max delay to verify your explanations **(5%)**.

Show the **post-SYNTHESIS** simulation results for the first 20 samples and the last 20 samples and indicate why your throughput is doubled **(10%)**. Show the errors of post-synthesis simulation outputs and Matlab floating-point results of direct-form FIR filter output $y[n]$ for $0 \leq n \leq 143$ **(10%)**. Show the area (resources) before and after speedup from the synthesis report **(10%)**.

5.1 Behavior Simulation

Experimental results demonstrate that after implementing the parallel processing architecture, the total time required to output the 144th sample $y[n]$ is reduced to **72 ns (from 8.5 ns to 80.5 ns)**. In contrast, the original architecture requires **144 ns (from 8.5 ns to 152.5 ns)** to complete the same operation. These data points confirm that the parallel processing architecture effectively halves the processing time, achieving a 2x speedup.

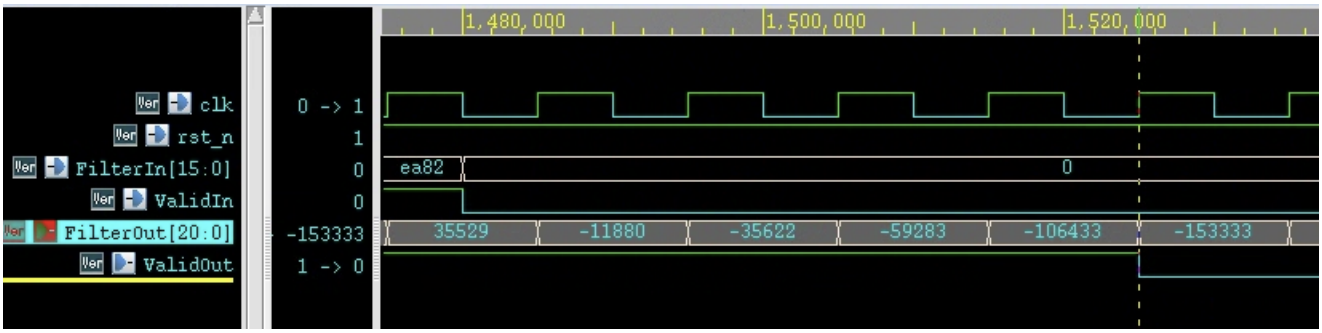

Figure 5.1: Original Behavior Simulation

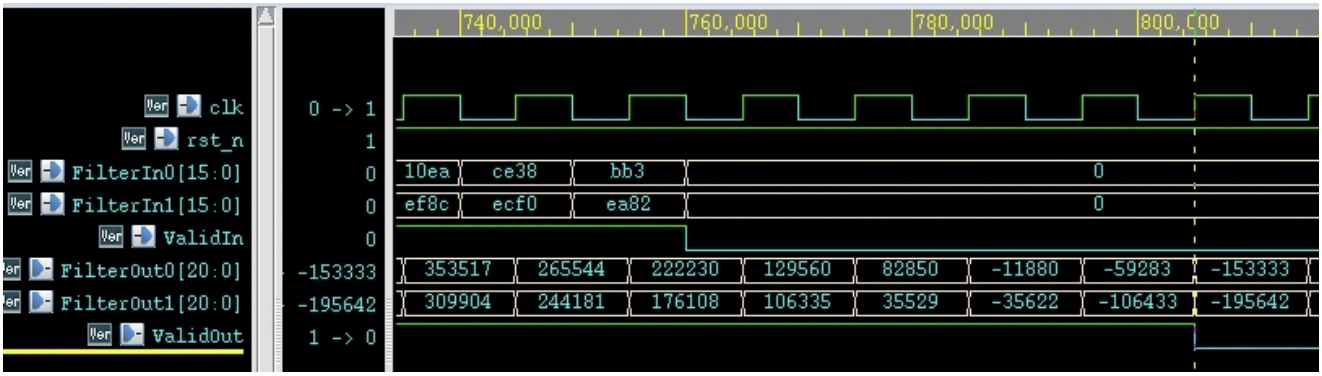

Figure 5.2: 2X Speedup Behavior Simulation

5.2 Critical Path Analysis & Timing Report

The design implements a pipeline stage at 5-tap intervals. Consequently, **the critical path delay consists of 1 multiplier and 4 adders**, significantly reducing the logic depth compared to the non-pipelined baseline.

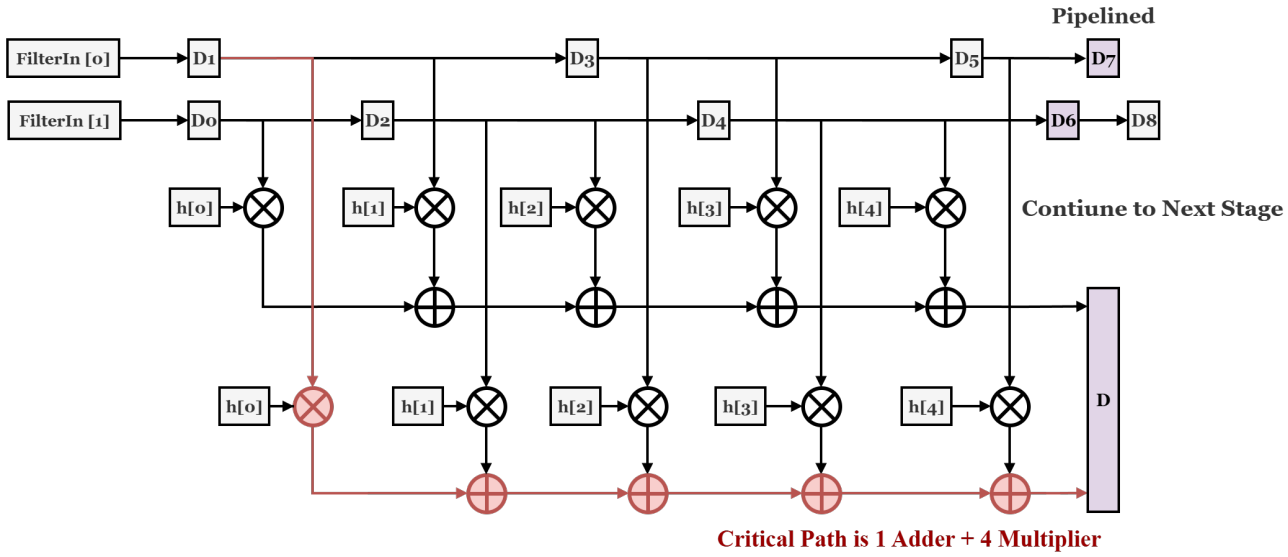

Figure 5.3: Block Diagram of 2X Speedup Filter Critical Path

The timing report validates that the critical path is strictly constrained between the successive launch register and capture register. The total combinational delay is measured at 1.38ns, which accounts for the logic depth of one multiplier and four adders and the timing constraint settings are the same as in Section 4.2.

```
Des/Clust/Port     Wire Load Model       Library
------------------------------------------------
Digital_Filter_Parallel
                   ZeroWireload          N16ADFP_StdCellff0p88v125c_ccs

Point                                    Incr       Path
-----------------------------------------------------------
clock clk (rise edge)                    0.00       0.00
clock network delay (ideal)              0.50       0.50
input external delay                     0.50       1.00 f
ValidIn (in)                             0.00       1.00 f
U7936/ZN (NR2D1BWP20P90)                 0.07       1.08 r
U7911/ZN (CKND2BWP20P90)                 0.11       1.19 f
U6212/ZN (CKND1BWP20P90)                 0.05       1.24 r
U7252/Z (BUFFD1BWP20P90)                 0.07       1.31 r
U7908/Z (BUFFD1BWP20P90)                 0.05       1.36 r
U1934/ZN (ND2D1BWP20P90)                 0.01       1.37 f
U1373/ZN (OAI21D1BWP20P90)               0.01       1.38 r
X_D_reg[20][0]/D (DFCNQD1BWP20P90)       0.00       1.38 r
data arrival time                                   1.38

clock clk (rise edge)                    1.00       1.00
clock network delay (ideal)              0.50       1.50
clock uncertainty                       -0.10       1.40
X_D_reg[20][0]/CP (DFCNQD1BWP20P90)      0.00       1.40 r
library setup time                      -0.02       1.38
data required time                                  1.38
-----------------------------------------------------------
data required time                                  1.38
data arrival time                                  -1.38
-----------------------------------------------------------
slack (MET)                                         0.00
```

Figure 5.4: Timing Report of 2X Speedup

5.3 Post-Synthesis Simulation and Error Analysis

Actually, the RMSE of Post-Synthesis is alos same as Behavior Simulation in Section 3.1 (only 0.00045324).

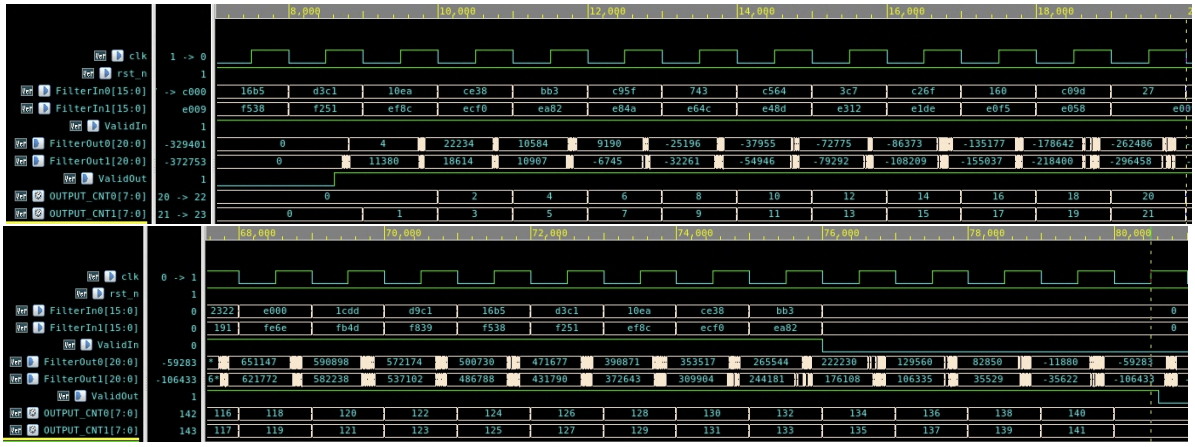

Figure 5.5: Post-Synthesis Timing Diagram (First 20 & Last 20)

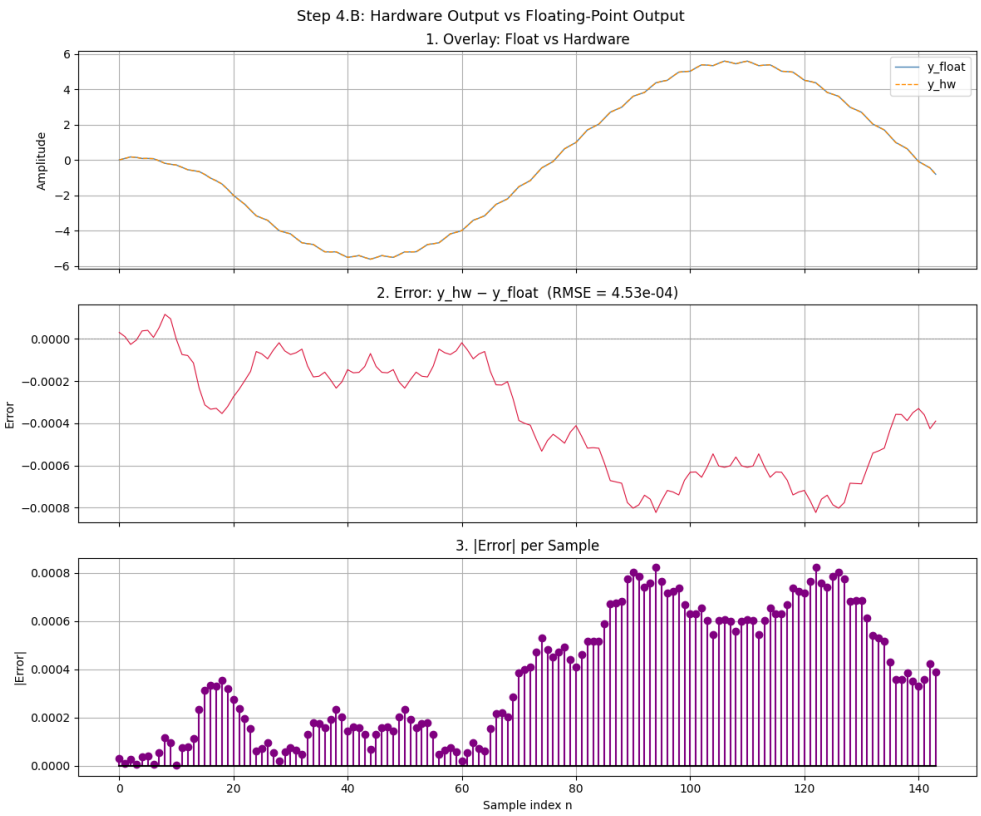

Figure 5.6: The errors of Post-Synthesis hardware outputs and Python floating-point result

5.4 Resources Comparison

Design	Total Area	Area Overhead
Original	2,943	1×
Two-Level Parallel	5,233	1.78×

Table 5.1: Area comparison before and after 2X speedup.